GLBL 570 · Methodologies of Global Studies · Module 4 · Week 8

GIS Workshop II: Lab Guide

Overlay · buffer · image classification · map and memo

60 minutes in class · QGIS 3.28 or later (ArcGIS Pro equivalents noted) · submit on Canvas

The Hour at a Glance

Time	Task	Data (folder)	What you keep
Before class	Unzip GLBL570_Week8_Lab.zip; install the SCP plug-in in QGIS	GLBL570_Week8_Lab.zip (Canvas)	—
0–15′	Task 1 · Overlay: how much forest is protected?	Task1_Overlay/	forest inside / outside protection, km²
15–30′	Task 2 · Buffer: which refugee camps are near conflict?	Task2_Buffer/	number of camps at 3 radii
30–45′	Task 3 · Image classification with SCP: summer, then summer + spring	Task3_Image_Classification/	accuracy of 2 runs + class map
45–55′	Task 4 · Make the map and write the memo	your outputs	1–2 maps (PNG/PDF) + memo
55–60′	Submit on Canvas	—	—
After class	Optional extensions: A spatial statistics · C OSINT aggregation	Extensions/	—

What is graded (1) exported map(s) of your analysis and (2) a 200–300-word memo that states your choices — which definition of "protected", which radius, which rule — defends them, and draws ONE policy implication. Optional: a short OSINT reflection and an AI reflection.

The most common weakness a memo that only lists the steps. Say what the pattern means and what it cannot show.

Week 8 Data: What Is New Since Week 7

	Week 7 · Fundamentals of GIS & Mapping	Week 8 · Spatial Analysis
Question	What is where? One layer, styled well	What do locations mean together? Layers combined, distances measured, patterns tested
Folder	lab_data/	GLBL570_Week8_Lab/ — one sub-folder per task; Week 7 files you need are copied in
Places	U.S. counties, world countries, Illinois, Chicago	Mount Hood – Columbia Gorge (6 counties, OR/WA) · Cox's Bazar refugee camps (Bangladesh) · contiguous U.S. counties
New layers	—	PAD-US 4.1 protected areas (3,043, with GAP status) · 33 Rohingya camp outlines · 410 UCDP events near the camps · ready-made training and check areas for classification
Rasters	NLCD and SRTM at 150 m for all of Illinois; one summer Sentinel-2 image	NLCD 2021 at its native 30 m (Tasks 1 and 3); a second, spring Sentinel-2 image (Task 3)
Coordinate systems	Mostly geographic (degrees); EPSG:5070 for display	Metres required: EPSG:5070 (U.S. Albers) for Task 1, EPSG:32646 (UTM 46N) for Task 2
Operations	Join · classify · symbolize · print layout	Intersect · difference · dissolve · buffer · select by location · count points · supervised classification (random forest) and accuracy assessment

Why new data: overlay needs two layers that overlap in the same place (forest and protected land), and buffering needs features and events measured in metres.

The Files You Will Use

File	Type · CRS as stored	Task	What it is
Task1_Overlay/focus_nlcd2021_30m.tif	raster, 30 m · EPSG:5070	1	NLCD 2021 land cover codes; forest = 41, 42, 43
Task1_Overlay/padus41_mthood_gorge.shp	3,043 polygons · EPSG:4326	1	Protected areas; field GAP_Sts = 1–4 (text)
Task1_Overlay/focus_counties_pnw.shp	6 polygons · EPSG:4269	1	Study-area outline
Task2_Buffer/coxsbazar_camps_2023.shp	33 polygons · EPSG:32646	2	Camp outlines (camp level only)
Task2_Buffer/ucdp_ged_2023_near_camps.csv	410 rows, longitude/latitude	2	Conflict events 2023; deaths_best, where_prec
Task3_Image_Classification/summer_2023/ (4 files)	1 band each, 10 m · EPSG:5070	3	Sentinel-2 Jun–Sep 2023: B02, B03, B04, B08 (× 10,000)
Task3_Image_Classification/spring_2023/ (4 files)	1 band each, 10 m · EPSG:5070	3	Sentinel-2 Apr–May 2023, same bands and grid
Task3_.../champaign_training_rois.gpkg	154 squares · EPSG:5070	3	Training areas, 5 classes, SCP import fields
Task3_.../champaign_validation_rois.gpkg	120 squares · EPSG:5070	3	Separate check areas for Accuracy
Task3_.../champaign_nlcd2021_30m.tif	raster, 30 m · EPSG:5070	3	NLCD 2021 for the same area (reference)
Extensions/ A_Spatial_Statistics · C_OSINT	counties · UCDP 2023 + countries	opt.	Optional work after class (last pages)

Keep every .csvt next to its .csv — it tells QGIS which columns are numbers. The README.md in the root folder lists every file.

Before You Start: Five-Minute Setup

1 Get the data

Download GLBL570_Week8_Lab.zip from Canvas and unzip it. Each task has its own folder; the README.md in the root folder lists every file.

2 Open QGIS 3.28 or later

Add layers by dragging files from the Browser panel. Find tools with the search box of the Processing Toolbox (Processing ▸ Toolbox).

3 Work in metres

Distances and areas are only right in a projected CRS. Task 1: Project ▸ Properties ▸ CRS = EPSG:5070. Task 2: EPSG:32646. Reproject layers before any buffer or area calculation.

4 Repair geometry first

PAD-US has a few invalid polygons after reprojection. Run Fix geometries before Intersection or Dissolve, or the tools stop with an error.

5 Install SCP before class

Task 3 needs the Semi-Automatic Classification Plugin (QGIS ▸ Plugins) and scikit-learn; the SCP installation page explains both. Extension A also needs GeoDa (geodacenter.github.io).

6 Save as you go

Save every output as a GeoPackage (.gpkg) in one folder, not as temporary layers — temporary layers disappear when QGIS closes.

Task 1 · Overlay: How Much Forest Is Protected? (1/2)

Question In six counties around Mount Hood (Oregon/Washington), what share of the forest lies inside protected areas — and how does the answer change with the definition of “protected”?

1 **Load and set the CRS.** Add focus_nlcd2021_30m.tif, padus41_mthood_gorge.shp, focus_counties_pnw.shp. Project ▸ Properties ▸ CRS ▸ EPSG:5070.

2 **Make a forest mask.** Raster ▸ Raster Calculator: ("focus_nlcd2021_30m@1" >= 41) AND ("focus_nlcd2021_30m@1" <= 43) → forest_mask.tif (1 = forest, 0 = not).

3 **Turn it into polygons.** Raster ▸ Conversion ▸ Polygonize (Raster to Vector) → field DN. Then Processing ▸ Extract by attribute: DN = 1 → forest.

4 **Prepare the protected layer.** Processing ▸ Reproject layer (PAD-US → EPSG:5070) ▸ Fix geometries ▸ Extract by expression: "GAP_Sts" IN ('1','2') — GAP_Sts is text, so use quotes.

5 **Dissolve it.** Vector ▸ Geoprocessing Tools ▸ Dissolve (no field) → protected_12. PAD-US records overlap; without Dissolve the area is counted twice.

What GAP status means

1 wilderness, left to natural processes

2 managed for biodiversity (refuges, wild rivers)

3 protected from development but open to logging, grazing, mining — most national-forest land

4 no known protection mandate

Strict definition = GAP 1–2. Broad definition = GAP 1–3. You will compute both.

Slow computer? First Raster ▸ Extraction ▸ Clip Raster by Mask Layer to one county (e.g. Hood River) and run the steps on that county only.

Task 1 · Overlay: Measure the Gap (2/2)

6 Protected forest. Vector ▸ Geoprocessing Tools ▸ Intersection: input forest, overlay protected_12 → forest_protected.

7 The gap. Vector ▸ Geoprocessing Tools ▸ Difference: input forest, overlay protected_12 → forest_gap.

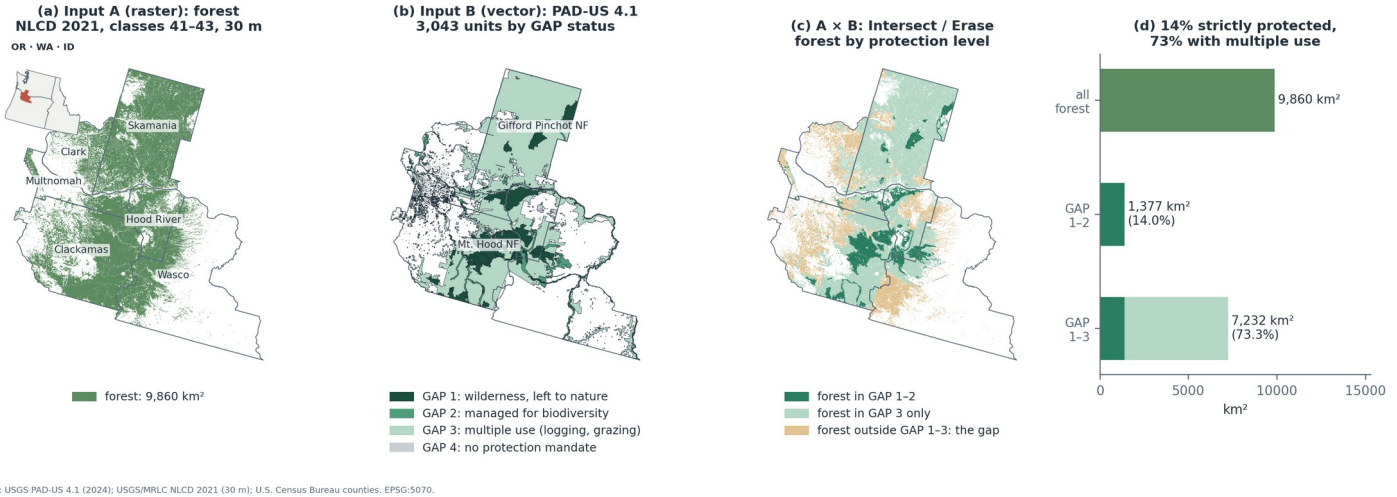
8 Measure. Open the Field Calculator on each layer, new decimal field km2 = area($geometry) / 1e6. Then Vector ▸ Analysis Tools ▸ Basic Statistics for Fields → SUM of km2.

9 Repeat with GAP 1–3. Steps 4–7 with "GAP_Sts" IN ('1','2','3'). Write both answers down.

Self-check (whole focus area)

All forest ≈ 9,860 km² · inside GAP 1–2 ≈ 1,377 km² (14.0 %) · inside GAP 1–3 ≈ 7,232 km² (73.3 %). Differences under 1 % are fine. A total far above 9,860 km² means you skipped Dissolve.

For the memo: which definition of "protected" do you use, and why?



What the result looks like (notebook figure): (a) forest, (b) PAD-US by GAP status, (c) the overlay, (d) the areas.

TASK 2 · 15–30′

Task 2 · Buffer: Which Refugee Camps Are Near Conflict?

1 **New project, CRS EPSG:32646.** Add coxsbazar_camps_2023.shp (33 camp outlines, already in UTM 46N, metres).

2 **Add the events as points.** Layer ▸ Add Layer ▸ Add Delimited Text Layer: ucdp_ged_2023_near_camps.csv, X field = longitude, Y field = latitude, CRS EPSG:4326.

3 **Reproject the events.** Processing ▸ Reproject layer → EPSG:32646. Buffering in degrees gives wrong distances.

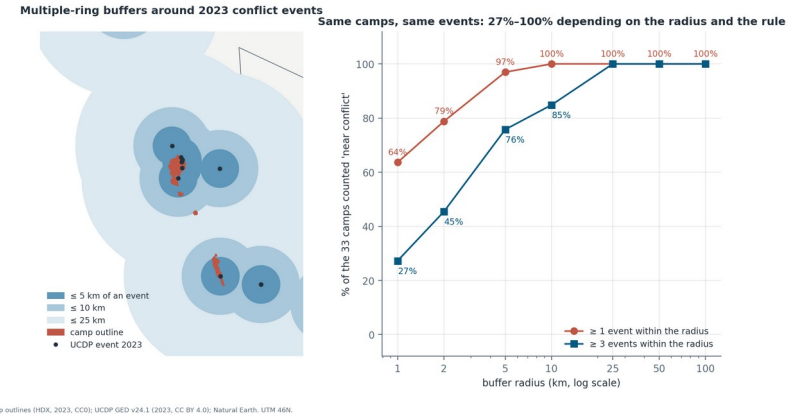
4 **Buffer the events.** Vector ▸ Geoprocessing Tools ▸ Buffer, distance 5000 m, tick Dissolve result → buf_5km. Repeat for 1 km and 10 km.

5 **Count camps inside.** Vector ▸ Research Tools ▸ Select by Location: select camps that intersect buf_5km. The number selected appears at the top of the attribute table.

6 **Stricter rule (≥ 3 events).** Buffer the camps by 5 km → Vector ▸ Analysis Tools ▸ Count Points in Polygon (points = events) → select NUMPOINTS >= 3.

Radius	Camps with ≥ 1 event	Camps with ≥ 3 events
1 km	21 of 33 (64 %)	9 (27 %)
2 km	26 (79 %)	15 (45 %)
5 km	32 (97 %)	25 (76 %)
10 km	33 (100 %)	28 (85 %)
25 km	33 (100 %)	33 (100 %)

Self-check: your counts should match this table.



For the memo: the answer runs from 27 % to 100 % depending on the radius and the rule. Pick one, say why, and show the range.

Ethics: these are camp outlines only. Never map individual shelters or blocks.

Task 3 · Classify a Satellite Image with SCP (1/2)

Question Can a random forest map land cover in Champaign–Urbana from a Sentinel-2 image — and how much does a second date help? Training and check areas are ready-made, so you only press the buttons and read the results.

1 **Open SCP.** Install the Semi-Automatic Classification Plugin before class (Plugins ▸ Manage and Install Plugins; random forest also needs scikit-learn — see the SCP installation page).

2 **Band set 1 = summer.** SCP ▸ Band set ▸ open file ▸ select the four files in Task3_Image_Classification/summer_2023/ (B02 blue, B03 green, B04 red, B08 near-infrared). In the working toolbar set RGB = 3-2-1 for natural colour, 4-3-2 for false colour.

3 **Load the training areas.** SCP ▸ Training input ▸ new file training.scpx ▸ Import vector: champaign_training_rois.gpkg. Match the fields macroclass_id, macroclass_name, class_id, class_name ▸ Import. You get 154 squares in 5 classes.

4 **Run 1.** SCP ▸ Classification ▸ input band set 1 ▸ tick Use Macroclass ID ▸ Algorithm: Random forest, number of trees 100 ▸ Run → class_summer.tif.

5 **Check it.** SCP ▸ Postprocessing ▸ Accuracy: classification = class_summer.tif, reference = champaign_validation_rois.gpkg, field macroclass_id ▸ Run. Write down the overall accuracy.

What goes into the model

Input (features): for every 10 m pixel, its reflectance in each band of the band set — 4 numbers in Run 1, 8 in Run 2.

Labels: the class of each training square: 1 Water · 2 Developed · 3 Forest · 4 Grass and pasture · 5 Cropland. Each square is 90 × 90 m (81 pixels) where NLCD 2021 shows one class only.

Check areas: 120 different squares, at least 300 m from any training square, used only for the accuracy report.

The model learns which reflectance goes with which class, then predicts every pixel of the image.

Task 3 · Add a Second Date and Compare (2/2)

6 **Band set 2 = summer + spring.** In the Band set tab add a new band set (+) and load the four summer files and the four files in spring_2023/ (8 bands).

7 **Run 2.** Classification with input band set 2, same settings → class_summer_spring.tif.

8 **Check it.** Postprocessing ▸ Accuracy with the same validation file. Compare with Run 1.

Self-check	Run 1 · summer	Run 2 · + spring
Overall accuracy	≈ 0.66	≈ 0.81
Water (recall)	0.65	0.79
Developed	0.76	0.87
Forest	0.68	0.71
Grass and pasture	0.62	0.74
Cropland	0.56	0.88

Random forests are random: ± 0.02 between runs is normal.

Why the second date helps

The summer image is a June–September median. In mid-summer, corn and soybean fields are dense green canopy and look almost like forest in these four bands: forest NDVI 0.83, cropland 0.80. Run 1 confuses them.

In April–May the fields are still bare soil while forest and grass are green. Adding that date lets the model separate them: cropland recall rises from 0.56 to 0.88.

Time is a feature. Choosing when the image was taken is as much an analytical decision as choosing the algorithm.

For the memo: report per-class accuracy, not just the overall number, and name the class your map gets wrong most often.

The labels come from NLCD 2021; the image is from 2023, so some “errors” are real change or label mistakes.

Task 4 · Make the Map and Write the Memo

The map (Project ▸ New Print Layout)

Check (lecture slide 5)	Ask yourself
1 Purpose	Can a reader say in one sentence what the map shows?
2 Visual variable	Darker = more? Categories in different hues?
3 Classification	Which method and how many classes? Stated?
4 Colour	Colour-blind safe; readable in greyscale?
5 Data integrity	Rates not counts; equal-area CRS; source and date on the map
6 Essentials only	Title, legend, scale, source; nothing decorative

Export: Layout ▸ Export as Image (PNG, 300 dpi) or Export as PDF. One map per analysis; at most two maps.

The memo (200–300 words)

Question — one sentence: what policy question does your analysis answer?

Data and units — layers, CRS, and the definitions you chose (GAP 1–2 or 1–3; radius; rule).

Operations and results — the tools in order, with your numbers.

One policy implication — what should change, for whom, on the strength of this result?

What it cannot establish — e.g. designation is not enforcement; straight-line distance is not travel distance; the map shows where, not why.

AI and ethics (optional) — did an AI suggestion help? Does any output locate vulnerable people precisely?

A template with these headings is in Task4_Memo/week8_memo_template.md.

Submit on Canvas: Checklist

	Item	Status
1	Map(s) from Task 1, 2 or 3, exported as PNG or PDF	Required
2	Memo, 200–300 words, with ONE policy implication	Required
3	OSINT reflection: how open-source data helps or risks humanitarian planning	Optional
4	AI reflection: did the AI's suggestion add value? What ethical concerns arose?	Optional

Name files lastname_week8_map1.png, lastname_week8_memo.pdf.

Due see Canvas.

Next week Module 5: Qualitative Coding & Reflexivity (MAXQDA).

What a strong memo does

States every choice (definition, radius, rule, statistic) and says why.

Reports numbers with units and the range they could take under another choice.

Draws one implication that follows from the numbers — no more.

Names one thing the analysis cannot show.

What a weak memo does

Lists the steps without saying what they mean.

Quotes one number as if it were the only possible answer.

If Something Goes Wrong

Symptom	Cause and fix
Layers do not line up	Different CRS without on-the-fly reprojection. Set the project CRS (EPSG:5070 or 32646) and reproject the layer.
A 5 km buffer covers the whole map	The layer is in degrees (EPSG:4326). Reproject to a metre-based CRS before buffering.
Protected area or forest total is far too large	Overlapping PAD-US records were counted twice. Dissolve before measuring area.
Intersection / Dissolve stops with an invalid-geometry error	Run Fix geometries on the input layer first.
Extract by expression returns nothing for GAP_Sts <= 2	GAP_Sts is text. Use "GAP_Sts" IN ('1','2').
CSV points do not appear	In Add Delimited Text Layer choose Point coordinates, X = longitude, Y = latitude, CRS EPSG:4326.
Polygonize or Intersection runs for many minutes	Clip the raster to one county first (Raster ▸ Extraction ▸ Clip Raster by Mask Layer).
SCP Import vector finds no classes	Match the four fields exactly: macroclass_id, macroclass_name, class_id, class_name.
SCP natural colour looks wrong	Bands are B02, B03, B04, B08 in that order: natural colour RGB = 3-2-1, false colour 4-3-2.

Extension A · Moran's I, LISA and Hot Spots

1 Keep the contiguous U.S. In QGIS: Processing ▸ Extract by expression on us_counties_acs2022.shp: NOT "state_abbr" IN ('AK','HI','PR') → save as counties48.shp.

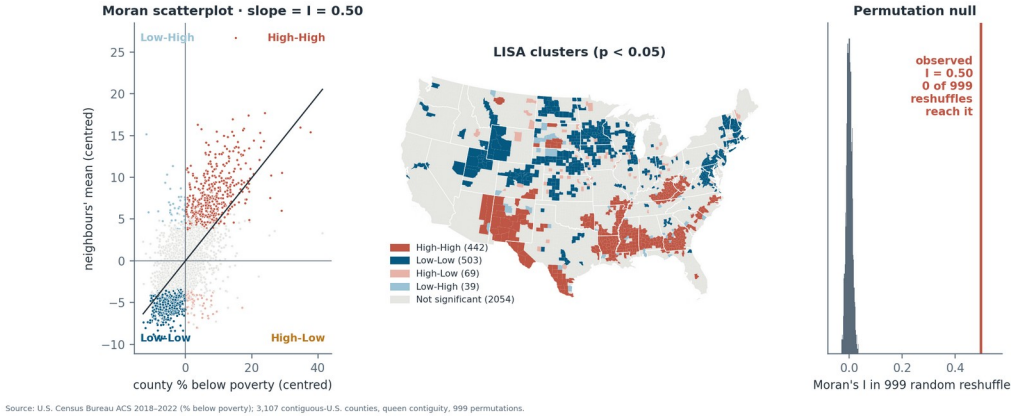
2 Open it in GeoDa. File ▸ Open ▸ counties48.shp.

3 Define neighbours. Tools ▸ Weights Manager ▸ Create ▸ ID variable geoid ▸ Queen contiguity ▸ save. GeoDa will report two counties with no neighbour (islands) — that is expected.

4 Global Moran's I. Space ▸ Univariate Moran's I ▸ variable pov_pct. Right-click the plot ▸ Randomization ▸ 999 permutations.

5 LISA. Space ▸ Univariate Local Moran's I ▸ tick Cluster Map.

6 Hot spots. Space ▸ Local G* ▸ Cluster Map.



Self-check Moran's I ≈ 0.50, pseudo p = 0.001. LISA: about 440 High-High and 500 Low-Low counties (counts vary slightly between runs). ArcGIS Pro: Spatial Autocorrelation (Global Moran's I) · Cluster and Outlier Analysis (Anselin Local Moran's I) · Hot Spot Analysis (Getis-Ord Gi*).

How to read it

Moran's I is the slope of the scatterplot: each county against the mean of its neighbours. 0 = no pattern; 0.50 = strong clustering.

LISA labels each county: High-High (poor among poor), Low-Low, and two kinds of outlier.

Gi* asks whether a county plus its neighbours is high or low as a whole — no outlier class.

For the memo: say which statistic you used and name one region it identifies. The map shows where, not why.

OPTIONAL · AFTER CLASS

Extension C · OSINT: From Event Points to a Country Map

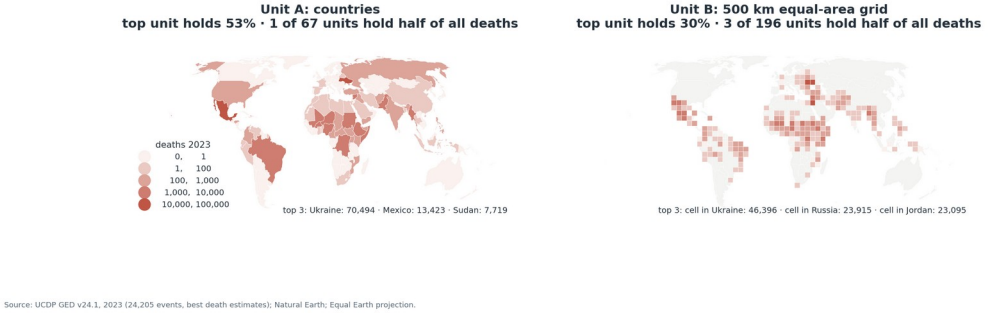
1 Add the events. Layer ▸ Add Layer ▸ Add Delimited Text Layer: Extensions/C_OSINT/ucdp_ged_events_2023.csv, X = longitude, Y = latitude, EPSG:4326 (24,205 points).

2 Add the countries. Extensions/C_OSINT/world_countries_ne110m.shp.

3 Sum deaths per country. Processing ▸ Join attributes by location (summary): join to world_countries, from the events, field deaths_best, summary sum → deaths_best_sum.

4 Map it. Symbology ▸ Graduated on deaths_best_sum; set the project CRS to EPSG:8857 (Equal Earth). Countries with no events stay a separate grey class.

5 Compare. Put the point map and the country map side by side.



Self-check the three highest totals are Ukraine (≈ 70,500 deaths), Mexico (≈ 13,400) and Sudan (≈ 7,700).

For the memo: what does the country map lose, and what risk does publishing exact points create?

Why aggregate?

The country map supports the same regional conclusions, but exact points also show where people were. Laid over camps or roads, they can help someone locate or target them.

The units change the answer: by country, one unit holds half the deaths; on a 500 km grid it takes three cells (the modifiable areal unit problem, lecture slide 41).

ACLED, named in the syllabus, needs a free registration; UCDP (CC BY 4.0) is used here instead.

Data Sources and Further Help

Data

PAD-US 4.1 — USGS Gap Analysis Project (2024), public domain.

NLCD 2021 — USGS / MRLC, public domain. SRTM — NASA/USGS, public domain.

Rohingya camp outlines — RRRC / ISCG Site Management Sector, UNHCR, IOM, via HDX (2023), CC0.

UCDP GED v24.1 — Uppsala Conflict Data Program (Sundberg & Melander 2013), CC BY 4.0.

ACS 2018–2022 county estimates and boundaries — U.S. Census Bureau.

Sentinel-2 L2A — Copernicus / ESA (2023).

Natural Earth 1:110m countries — public domain.

Further help

The companion notebook GLBL570_Week8_Spatial_Analysis_Lab.ipynb runs every task in Python and prints the numbers used for the self-checks.

QGIS user manual: docs.qgis.org — Processing algorithms reference.

GeoDa workbook: geodacenter.github.io/documentation.html.

SCP v8 manual: semiautomaticclassificationmanual-v8.readthedocs.io.

README.md in the lab folder lists every file; the course repository's data READMEs list every field, CRS and known pitfall.